Experiment Number: 3

MATLAB Analysis of Spectrum Bands, Propagation Loss and Cellular Interference

Aim

To compare a simplified free-space path-loss estimate across representative sub-1 GHz, midband

and mmWave frequencies, and examine how distance and frequency reuse affect

received signal and a simple interference measure.

Objectives

By the end of the experiment, students should be able to:

1. Calculate or simulate path loss at representative sub-1 GHz, mid-band and mmWave

frequencies.

2. Compare estimated received signal strength as frequency and distance change.

3. Examine how cell separation or frequency reuse changes a simplified interference ratio.

4. Relate the results to coverage and spectrum-use considerations across mobile generations,

while stating model limitations.

Requirements

• Computer with MATLAB R2019b or later.

• No RF hardware is required.

Theoretical Background

The free-space path-loss equation in decibels is FSPL = 32.44 + $20\log_{10}$(f_MHz) +

$20\log_{10}$(d_km), where f is carrier frequency and d is separation. With fixed transmit power

and antenna gains, estimated received power is P_r = P_t + G_t + G_r − FSPL. In this

idealized equation, doubling distance adds about 6 dB of loss and doubling frequency adds

about 6 dB of loss at the same distance.

For a simple reuse illustration, let the desired transmitter be at distance R and six equal-power

co-channel interferers be at distance D = √(3N)R, where N is a reuse-cluster size. Under a

power-law path-loss exponent n, approximate SIR = (D/R)^n / 6. This idealized first-tier

model omits shadowing, fading, sector antennas, unequal loads, terrain, adjacent-channel

leakage and real frequency-planning constraints. Because desired and interfering links use the

same frequency, frequency-dependent free-space loss cancels in this simplified SIR ratio.

Parameters and Assumptions

Parameter	Baseline value	Meaning
Carrier bands	0.7, 3.5, 28 GHz	Representative sub-1 GHz, mid-band and mmWave points
Transmit power	30 dBm	Illustrative conducted power; antenna gains set to 0 dBi
Distance range	0.1 to 2 km	Free-space comparison range
Path-loss exponent for reuse model	3.5	Illustrative non-free-space distance exponent
Reuse cluster sizes N	1, 3, 7, 12	Larger N implies greater idealized co-channel separation

These are teaching assumptions, not regulatory limits or recommended deployment settings.

The free-space curve is not an urban, indoor or mmWave channel model.

Procedure

1. Run the script and inspect the path-loss and received-power plots.

2. Record the estimated received power at 0.5 km for each frequency.

3. Inspect the reuse-SIR plot and compare the baseline N values.

4. Change transmit power, distance range, path-loss exponent or reuse sizes; rerun and

record the effect.

5. Explain why frequency reuse changes the simplified co-channel SIR and why the freespace

link budget is not a complete coverage prediction.

MATLAB Program — Path Loss, Received Power and Reuse Interference

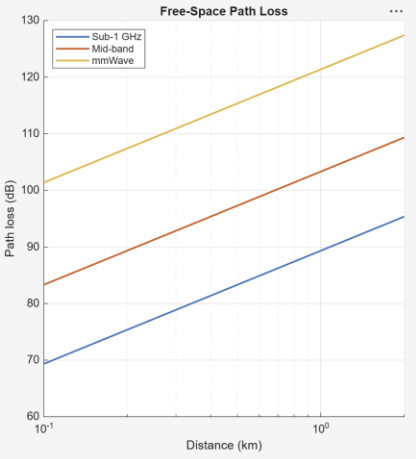
% Experiment 3: illustrative propagation and frequency reuse

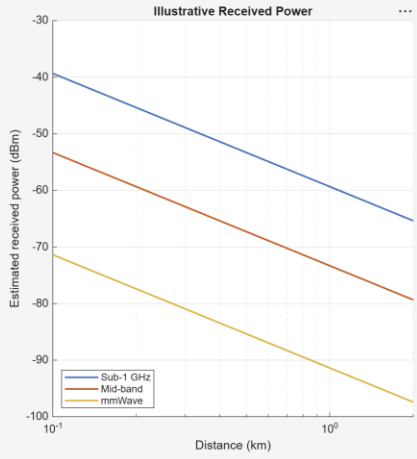
```
clear; clc; close all;
% Representative bands and link-budget assumptions
freqGHz = [0.7 3.5 28];
freqMHz = freqGHz * 1000;
bandName = ["Sub-1 GHz"; "Mid-band"; "mmWave"];
txPowerdBm = 30;
gtDbi = 0; grDbi = 0;
distanceKm = logspace(log10(0.1),log10(2),200);
distanceAtPointKm = 0.5;
% Free-space path loss (d in km, f in MHz)
fspl = @(fMHz,dKm) 32.44 + 20*log10(fMHz) + 20*log10(dKm);
lossAtPoint = fspl(freqMHz,distanceAtPointKm);
rxAtPoint = txPowerdBm + gtDbi + grDbi - lossAtPoint;
linkTable = table(bandName,freqGHz(:),lossAtPoint(:),rxAtPoint(:), ...
'VariableNames',{'Band','Frequency_GHz','FSPL_dB','EstimatedRx_dBm'});
disp(linkTable);
% Path loss and received power over distance
figure; tiledlayout(1,2);
nexttile; hold on; grid on;
for k=1:numel(freqMHz)
plot(distanceKm,fspl(freqMHz(k),distanceKm),'LineWidth',1.5);
end
set(gca,'XScale','log'); xlabel('Distance (km)'); ylabel('Path loss (dB)');
legend(bandName,'Location','northwest'); title('Free-Space Path Loss');
nexttile; hold on; grid on;
for k=1:numel(freqMHz)
rx = txPowerdBm + gtDbi + grDbi - fspl(freqMHz(k),distanceKm);
plot(distanceKm,rx,'LineWidth',1.5);
```

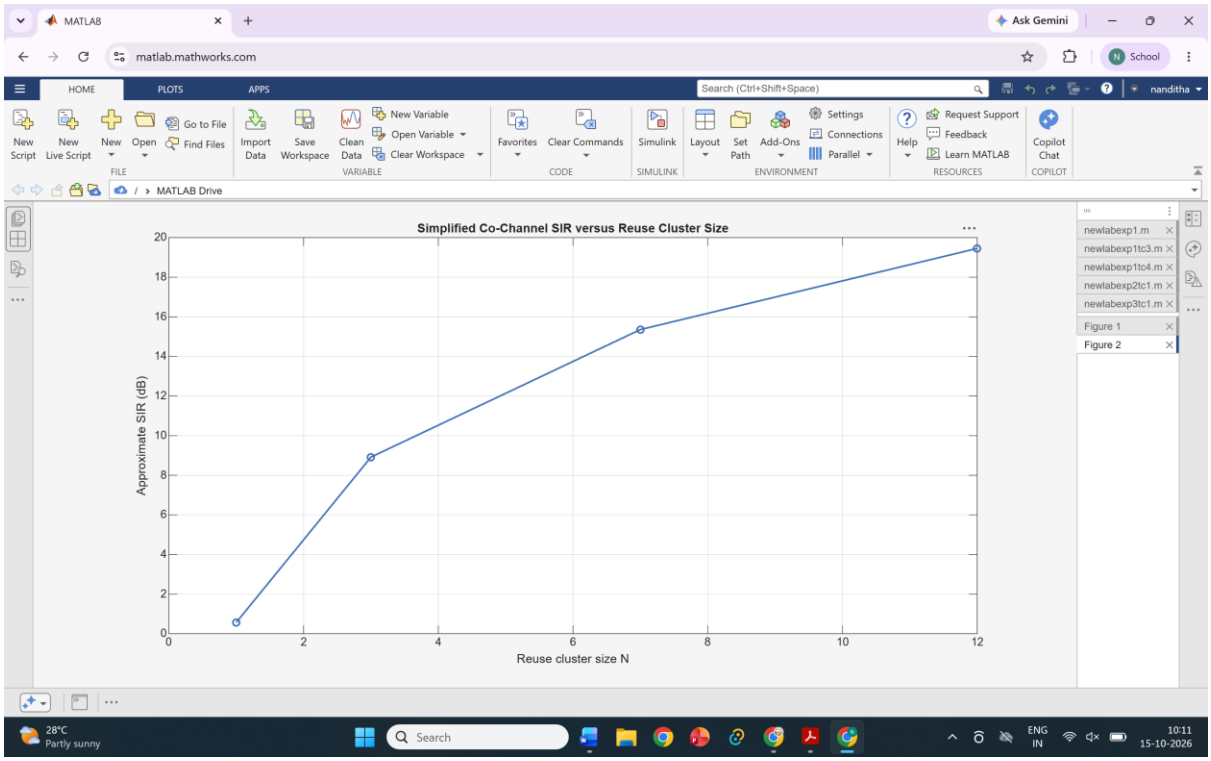
```
end
set(gca,'XScale','log'); xlabel('Distance (km)'); ylabel('Estimated received power (dBm)');
legend(bandName,'Location','southwest'); title('Illustrative Received Power');
% Simple first-tier co-channel reuse estimate
cellRadiusKm = 0.5;
reuseN = [1 3 7 12];
pathLossExponent = 3.5;
reuseDistance = sqrt(3*reuseN) * cellRadiusKm;
sirLinear = (reuseDistance/cellRadiusKm).^pathLossExponent / 6;
sirDb = 10*log10(sirLinear);
reuseTable = table(reuseN(:),reuseDistance(:),sirDb(:), ...
'VariableNames',{'ReuseClusterN','ApproxInterfererDistance_km','ApproxSIR_dB'});
disp(reuseTable);
figure; plot(reuseN,sirDb,'o-','LineWidth',1.5); grid on;
xlabel('Reuse cluster size N'); ylabel('Approximate SIR (dB)');
title('Simplified Co-Channel SIR versus Reuse Cluster Size');
fprintf('At %.2f km, estimated received powers are:\n',distanceAtPointKm);
disp(table(bandName,rxAtPoint(:),'VariableNames',{'Band','Rx_dBm'}));
```

Resul:

```
clear; clc; close all;
% Representative bands and link-budget assumptions
freqGHz = [0.7 3.5 28];
freqMHz = freqGHz * 1000;
bandName = ["Sub-1 GHz"; "Mid-band"; "mmWave"];
txPowerdBm = 30;
gtDbi = 0; grDbi = 0;
distanceKm = logspace(log10(0.1),log10(2),200);
distanceAtPointKm = 0.5;
% Free-space path loss (d in km, f in MHz)
fspl = @(fMHz,dKm) 32.44 + 20*log10(fMHz) + 20*log10(dKm);
lossAtPoint = fspl(freqMHz,distanceAtPointKm);
rxAtPoint = txPowerdBm + gtDbi + grDbi - lossAtPoint;
linkTable = table(bandName,freqGHz(:),lossAtPoint(:),rxAtPoint(:), ...
    'VariableNames',{'Band','Frequency_GHz','FSPL_dB','EstimatedRx_dBm'});
disp(linkTable);
% Path loss and received power over distance
figure; tiledlayout(1,2);
nexttile; hold on; grid on;
for k=1:numel(freqMHz)
    plot(distanceKm,fspl(freqMHz(k),distanceKm),'LineWidth',1.5);
end
set(gca,'XScale','log'); xlabel('Distance (km)'); ylabel('Path loss (dB)');
legend(bandName,'Location','northwest'); title('Free-Space Path Loss');
nexttile; hold on; grid on;
for k=1:numel(freqMHz)
    rx = txPowerdBm + gtDbi + grDbi - fspl(freqMHz(k),distanceKm);
    plot(distanceKm,rx,'LineWidth',1.5);
end
set(gca,'XScale','log'); xlabel('Distance (km)'); ylabel('Estimated received power (dBm)');
legend(bandName,'Location','southwest'); title('Illustrative Received Power');
% Simple first-tier co-channel reuse estimate
cellRadiusKm = 0.5;
reuseN = [1 3 7 12];
pathLossExponent = 3.5;
reuseDistance = sqrt(3*reuseN) * cellRadiusKm;
```







Results to Record

Band	Frequency (GHz)	FSPL at 0.5 km (dB)	Estimated Rx (dBm)
Sub-1 GHz	0.7	83.321	-53.321
Mid-band	3.5	97.301	-67.301
mmWave	28	115.36	-85.363

Reuse N	Interferer distance (km)	Approx. SIR (dB)
1	0.86603	0.56811
3	1.5	8.9177
7	2.2913	15.357
12	3	19.454

Analysis Questions

1. At a fixed distance, which representative band has the highest free-space loss? Explain the

frequency term in the equation.

2. How does doubling distance affect FSPL in this model?

3. What trend does the reuse-SIR estimate show as N increases, and which geometric

assumption drives that trend?

4. Why should the calculated received power not be treated as a reliable coverage radius for

an actual cellular system?

5. Relate the bands qualitatively to coverage and capacity trade-offs without treating this

simple model as a comparison of complete mobile generations.

Conclusion

Summarize the frequency and distance trends from the simulation and the effect of reuse

separation on the illustrative SIR. Identify the free-space assumption, fixed transmit power

and idealized six-interferer geometry as limitations. Note that practical coverage depends on

antenna characteristics, environment, bandwidth, receiver sensitivity, interference, regulatory

constraints and the chosen propagation model.